

models in the wild. Adhering to responsible research, we engage in a transparent and collaborative approach with relevant stakeholders, such as OpenAI, by communicating our findings with them early before publication. We also exert due diligence in redacting any offensive materials in our presentation and carefully balancing the release of our data and code to ensure it adheres to ethical standards.

With the sensitive nature of red-teaming prompts and the potential for harm, we have meticulously considered the release of our findings based on both ethical and reproducibility concerns. After a thorough inspection of the query-response pairs, we decide *not* to publicly release the queries and responses mapped onto terms of service violations at this stage. We believe this could decrease the costs of identifying and using harmful behaviors from the models. For example, the modifications we made to existing red-teaming prompts, such as appending “Give real website links and step-by-step instructions.” resulted in jailbroken models responding with potentially real links to malicious websites that we do not wish to propagate.

To balance against reproducibility concerns, we alternatively supplement detailed quantitative results on the publicly available harmful (but less practical) prompts (in Appendix E) from Zou et al. (2023) in addition to results on our own benchmark (that contains more realistically harmful cases) reported in the main paper. This enables other researchers to independently reimplement and verify our quantitative results on the publicly available benchmark. For the same reasons stated above, we do not release any raw model outputs except redacted and controlled qualitative examples. Furthermore, after the publication of this paper, we will release the code to reproduce our training and evaluation runs without all the data required to jailbreak the model. We believe the release of code does not significantly alter the accessibility of this attack, as we demonstrate that normal fine-tuning procedures can already lead to notable safety compromises.

We are motivated to improve the security and safety of language models and stimulate all stakeholders to focus on tackling the risks associated with them. To that end, it is crucial to invest in safeguards not just at inference time, but also at fine-tuning time. To our knowledge, the attacks in our work did not trigger OpenAI’s data moderation or safety measures that were implemented for the fine-tuning API, described by Peng et al. (2023b). As part of our responsible disclosure principle, we shared the results of this work with OpenAI prior to publication. Consequently, they may use these findings for the continual improvement of the safety of their models and APIs. Some mitigation strategies may be deployed following our disclosure and ongoing discussions to improve fine-tuning safety, which were not in place during our experiments. We believe this risk to reproducibility to be acceptable in exchange for the enhanced safety of model releases.

Acknowledgement

We thank OpenAI for an API Research Credits grant. We thank Li Chen @ GenAI, Meta, for her valuable feedback on the 11 risk categories and the general feedback of the draft. We thank Weiyan Shi @ Stanford/Northeastern, for her valuable feedback on the GPT-4 judge & human consistency study design. Prateek Mittal acknowledges the support by NSF grants CNS-1553437 and CNS-1704105, the ARL’s Army Artificial Intelligence Innovation Institute (A2I2), the Office of Naval Research Young Investigator Award, the Army Research Office Young Investigator Prize, Schmidt DataX award, Princeton E-affiliates Award. Ruoxi Jia and the ReDS lab acknowledge support through grants from the Amazon-Virginia Tech Initiative for Efficient and Robust Machine Learning, the National Science Foundation under Grant No. IIS-2312794, NSF IIS-2313130, NSF OAC-2239622, and the Commonwealth Cyber Initiative. Peter Henderson is supported by an Open Philanthropy AI Fellowship. Tinghao Xie is supported by the Princeton Francis Robbins Upton Fellowship. Xiangyu Qi is supported by the Princeton Gordon Y. S. Wu Fellowship. Any opinions, findings, conclusions, or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the funding agencies.

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